

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			most reactive 1 Metal C 2 Metal B 3 Copper 4 Metal A 5 Metal D award (2) for all four correct award (1) if C and B are above copper and A and D are below copper but the order is wrong			2	2		2
	(b)	(i)		(metal) Z → (metal) W		1		1		1
		(ii)		0.1			1	1		1
	(c)			the <u>copper ions</u> / Cu ²⁺ ions <u>gain (2) electrons</u> ≡ <u>reduction</u> (1) the <u>zinc atoms</u> / Zn <u>loses (2) electrons</u> ≡ <u>oxidation</u> (1) award (1) for general explanation of both oxidation and reduction without reference to the equation reference to gaining oxygen – neutral		2		2		2
				Question 3 total	0	3	3	6	0	6